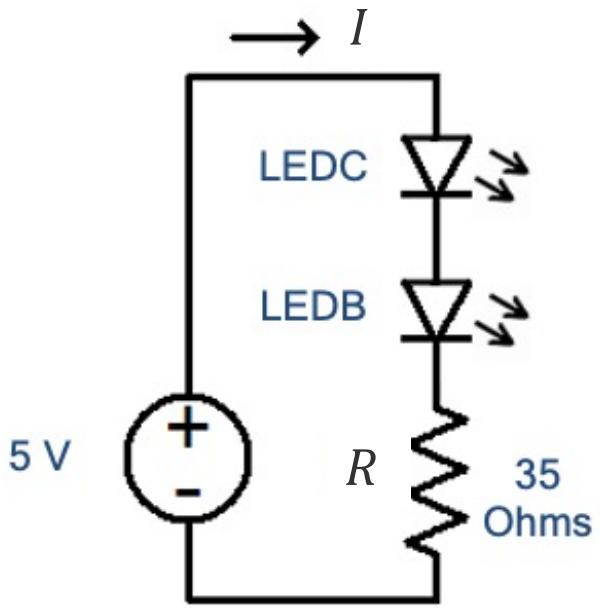
Light Emitting Diodes

The following subjects will be covered

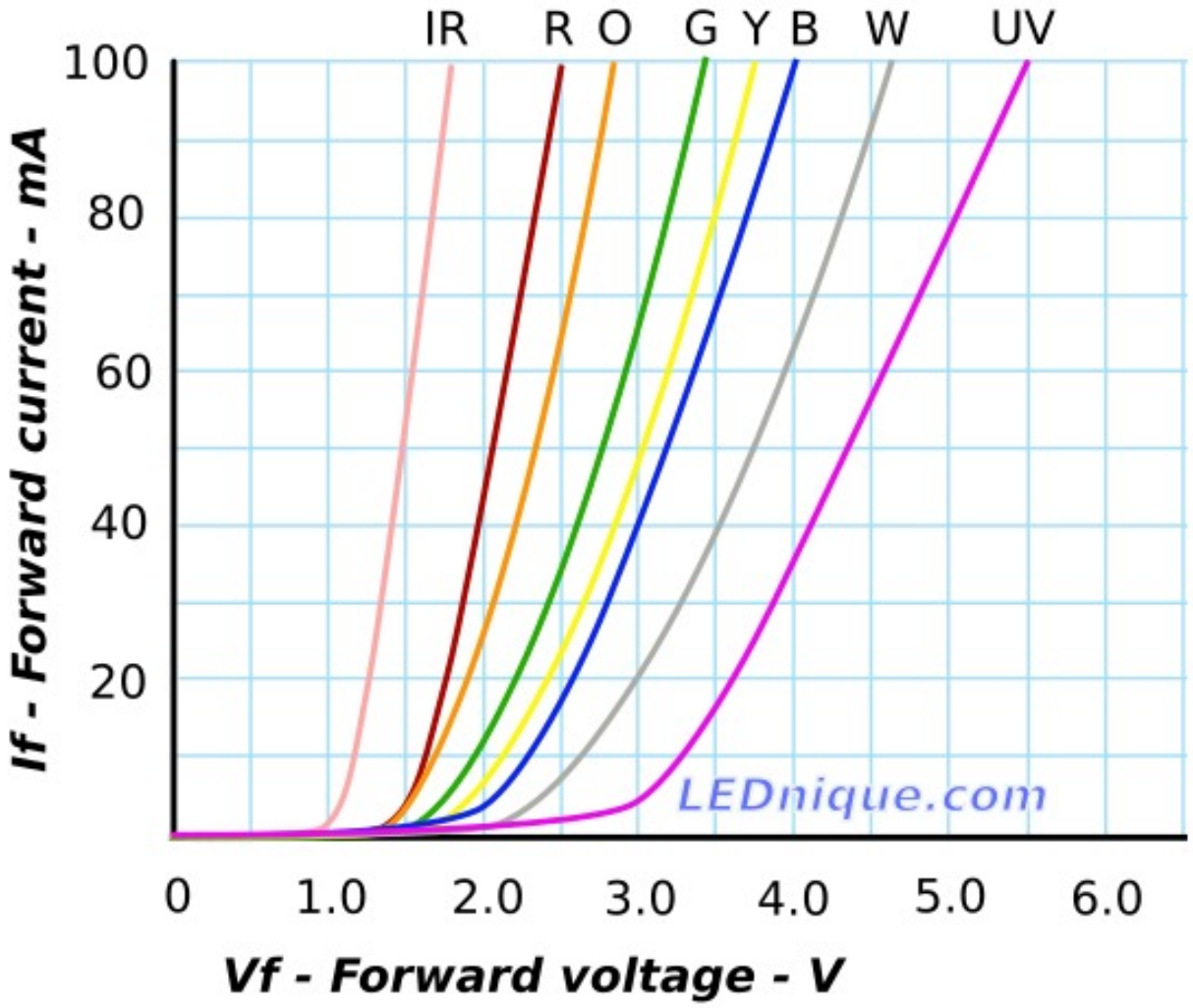
- LED Circuits.

Examples

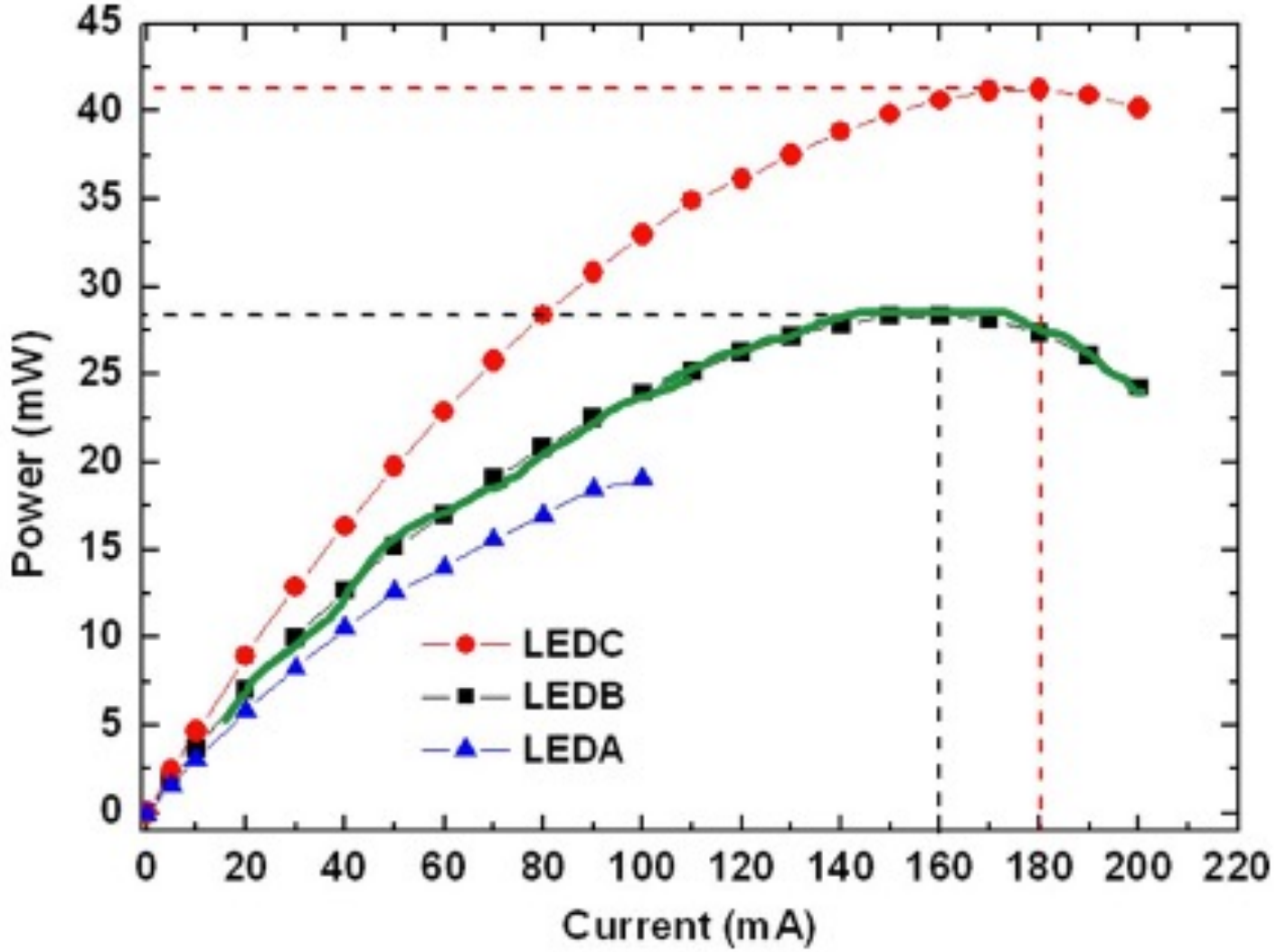
For the given LED circuit, find the luminous flux of each LED. The PI curves of the diodes are as shown.



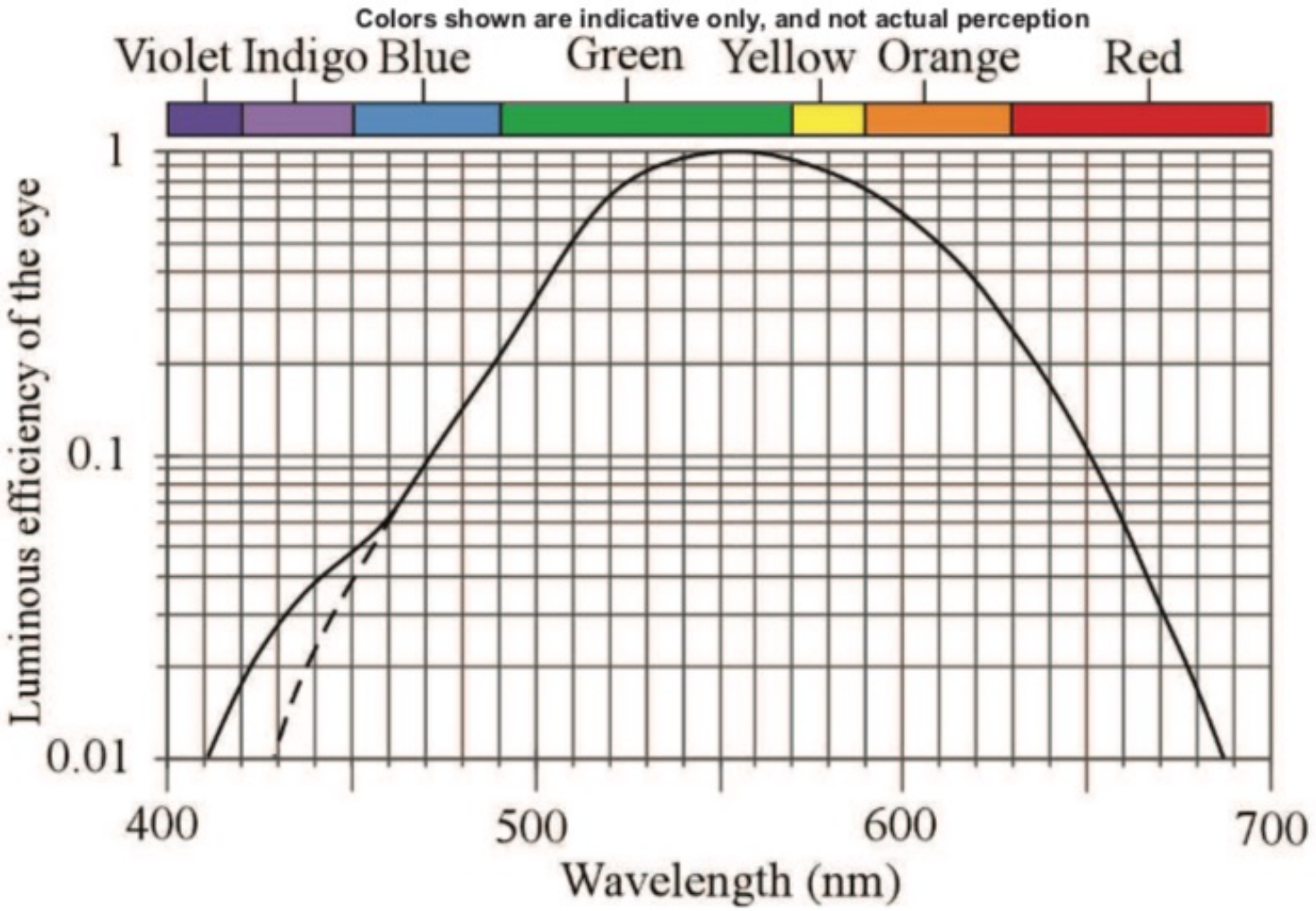
Examples


IR
R
O
G
Y
B
W
UV
100
80
60
40
20
If - Forward current - mA
LEDnique.com
0
1.0
2.0
3.0
4.0
5.0
6.0
Vf - Forward voltage - V

Examples

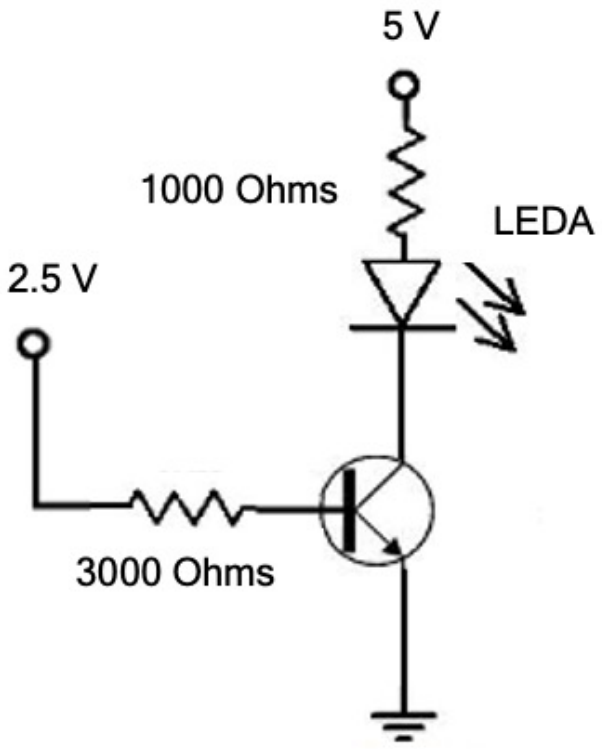

Power (mW)
Current (mA)
LEDC
LEDB
LEDA
0
5
10
15
20
25
30
35
40
45
0
20
40
60
80
100
120
140
160
180
200
220

Examples


Colors shown are indicative only, and not actual perception
Violet Indigo Blue Green Yellow Orange Red
1
0.1
0.01
Luminous efficiency of the eye
400
500
600
700
Wavelength (nm)

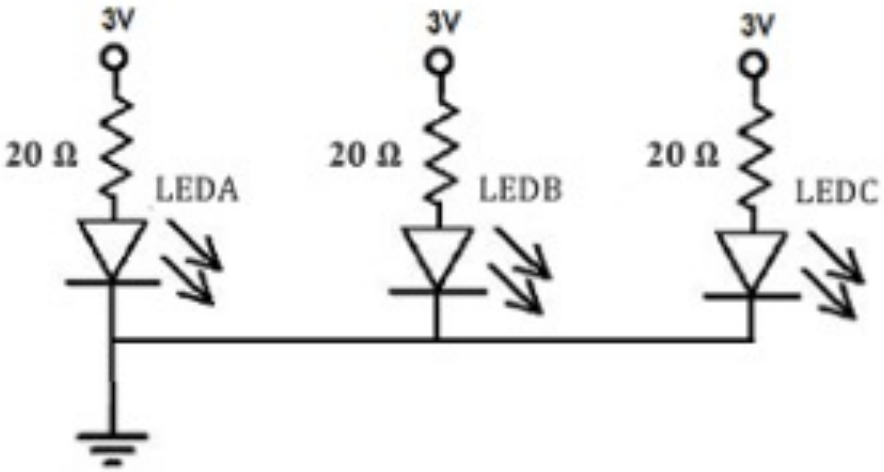
Examples

Find the luminous flux of the LED in the circuit below. The transistor has β=120. Use the given PI curves.



Examples

Given the circuit below, find the color mix percentages for each of the red, green and blue LEDs.



Examples

A light color mix of 40% red, 20% green and 40% blue is required. Design a circuit that uses the LEDs given in the figure below to produce the required mix. The design should use a +5V source.

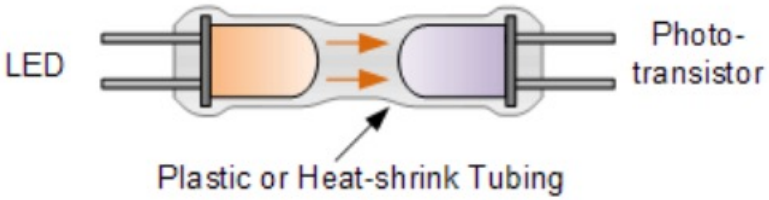
Opto-couplers and Opto-isolators

The following subjects will be covered

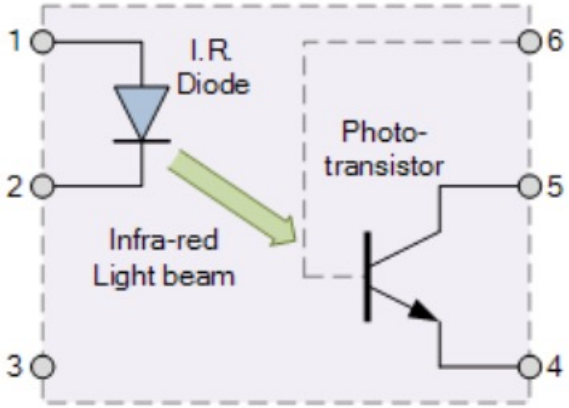
- Definition and Operation.
- Popular Opto-coupler ICs.
- Applications.

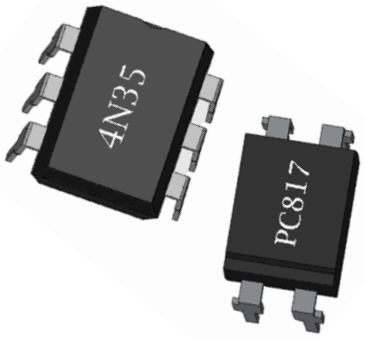
Opto-couplers and Opto-isolators

Similar to transformers that provide "electrical isolation" between the higher voltage on the primary side and the lower voltage on the secondary side, opto-couplers provide "electrical isolation" between an input source and an output load using light.



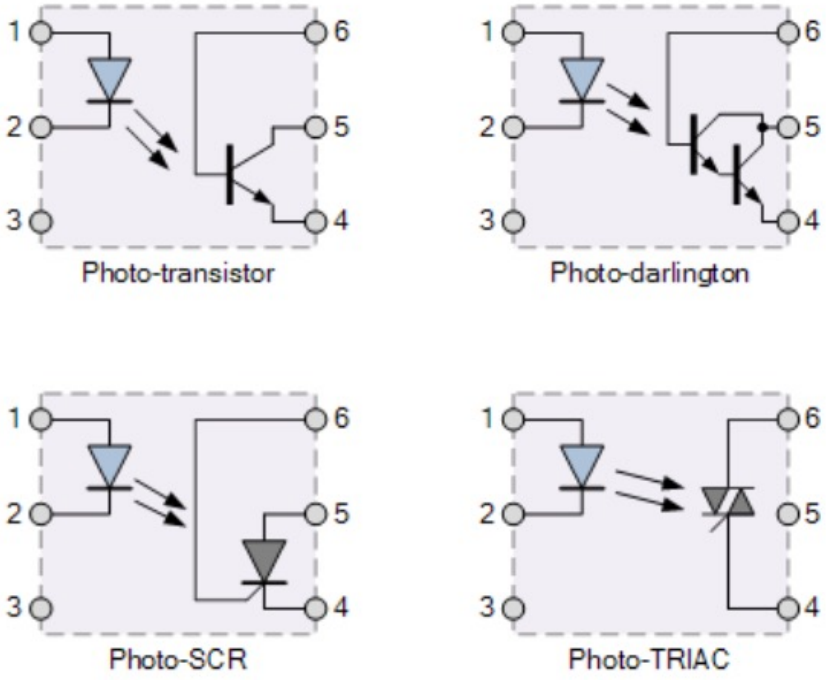
They allow systems such as power circuits to be controlled from a PC's output port, microcontrollers, digital switches or from a low voltage data signal (digital circuits).





Opto-couplers Types

Opto-couplers and opto-isolators come in four basic types. The use of a particular type depends on applications.

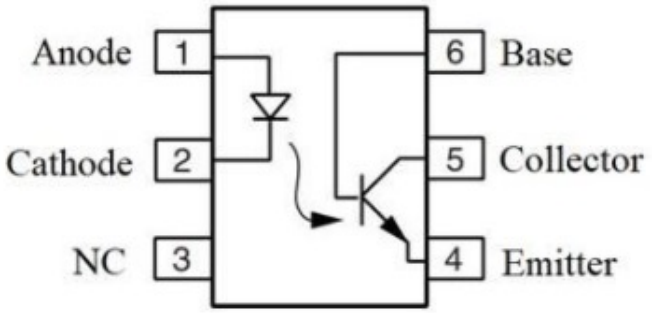


The main advantage of opto-couplers is their high electrical isolation between the input and output terminals allowing relatively small digital signals to control much largeer AC voltages, currents and power.

Popular Opto-couplers ICs

The 4N25, 4N26, 4N27 and 4N28 devices consist of a gallium arsenide infrared emitting diode optically coupled to a monolithic silicon phototransistor detector.

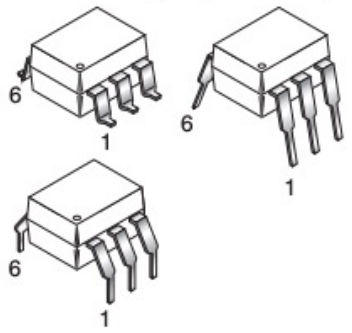




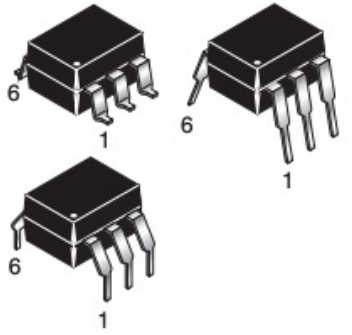
Popular Opto-couplers ICs

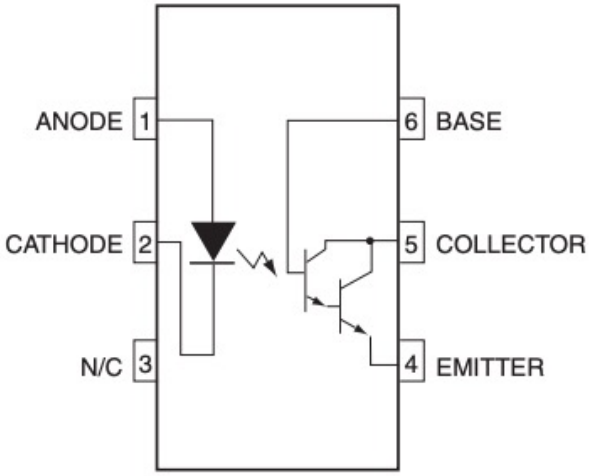
The 4N29, 4N30, 4N31, 4N32 and 4N33 devices consist of a gallium arsenide infrared emitting diode optically coupled to a monolithic silicon photo-Darlington detector.

White Package (-M Suffix)



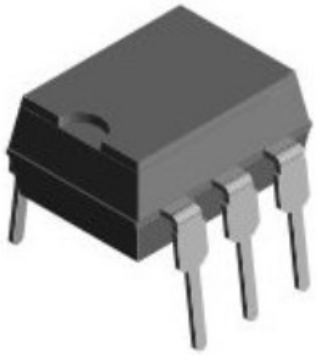
Black Package (No -M Suffix)

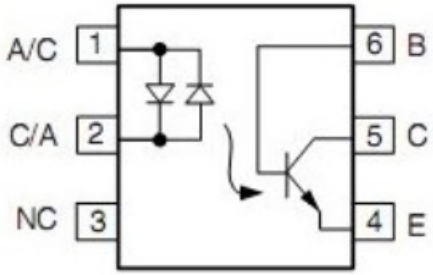




Popular Opto-couplers ICs

The H11AA1 is a bi-directional input optically coupled isolator consisting of two inverse parallel gallium arsenide infrared LEDs coupled to a silicon NPN phototransistor.

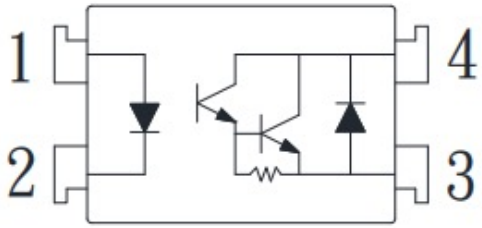




Popular Opto-couplers ICs

The THT KP40101B provides High isolation voltage (up to 5000 V) between input and output, with high current transfer ratio.

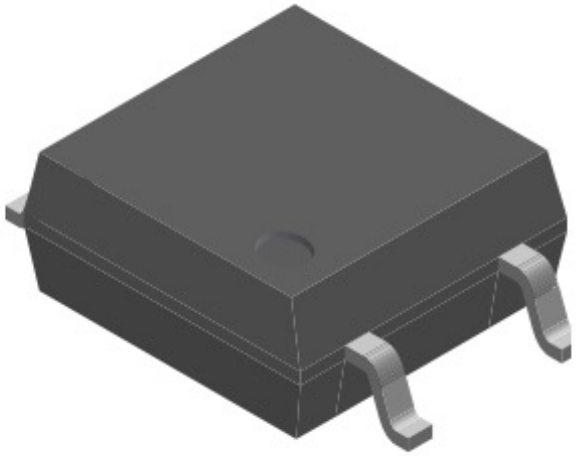


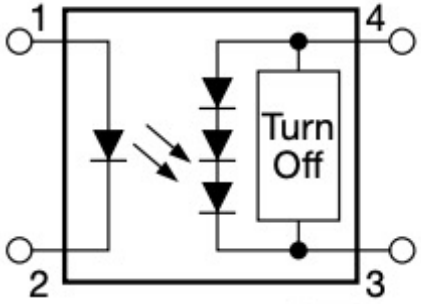


1. Anode
2. Cathode
3. Emitter
4. Collector

Popular Opto-couplers ICs

The VOM1271 is a stand-alone optically isolated MOSFET driver.





Popular Opto-couplers ICs

The FOD852 is a 4-Pin High Operating Temperature Photodarlington Optocoupler.

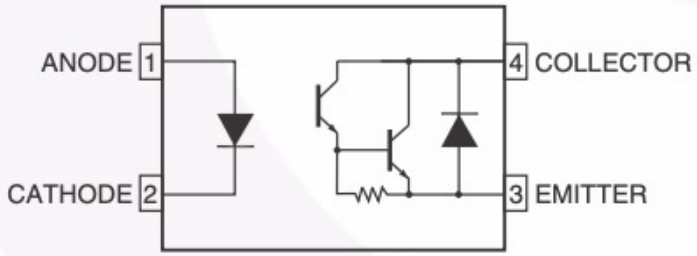


Figure 1. Schematic

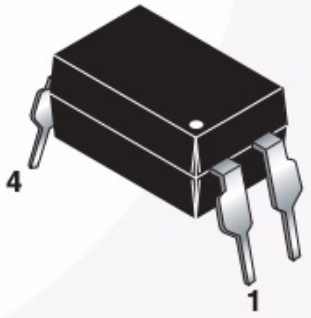
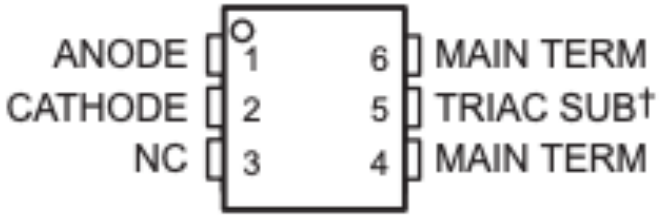


Figure 2. Package Outlines

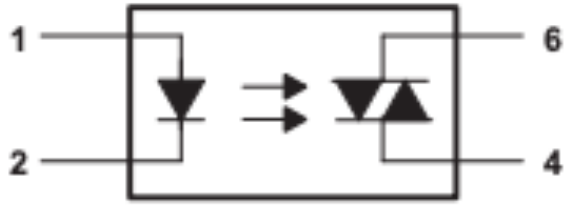
Popular Opto-couplers ICs

The MOC3020 is a Gallium-Arsenide-Diode Infrared Source and Optically-Coupled Silicon Traic Driver. Output Driver Designed for 220 Vac.



† Do not connect this terminal
NC – No internal connection

logic diagram



Applications

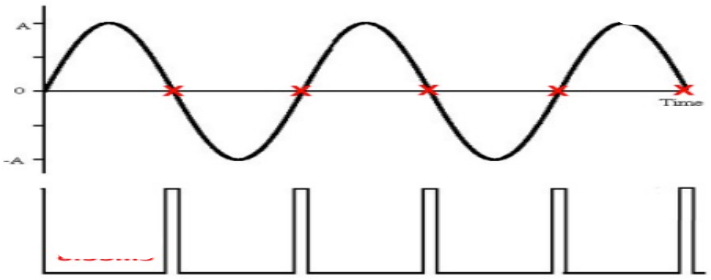
The main application of optocoupler is to isolate two different electrical circuits.

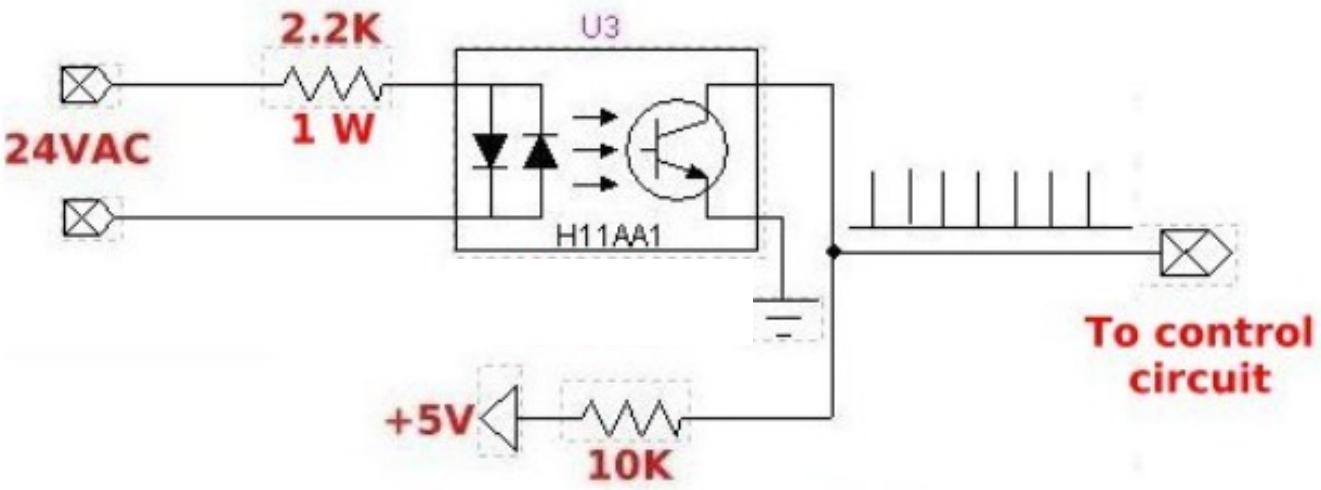
Other applications include:

- General Purpose Switching Circuits
- Interfacing and coupling systems of different potentials and impedances
- I/O Interfacing
- Solid State Relays
- Power supply regulators
- Digital logic inputs
- Microprocessor inputs
- Ground isolation.
- High voltage monitoring and sensing circuits.
- Lamp dimmer circuits.
- Motor and lighting control circuits.
- Switched-mode power supply (SMPS).
- DC to DC converters circuits in automobiles.

Zero-crossing Detector

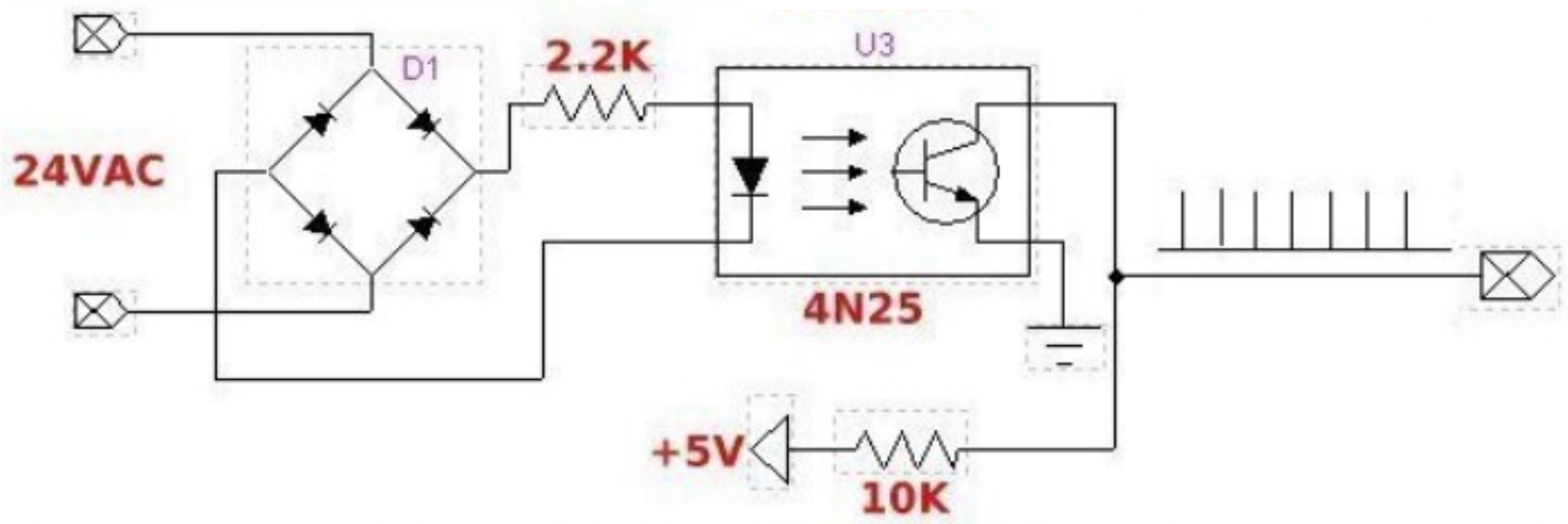
A zero-crossing detector is used to generate a sync pulse related to the AC voltage phase angle often used in power control circuits.





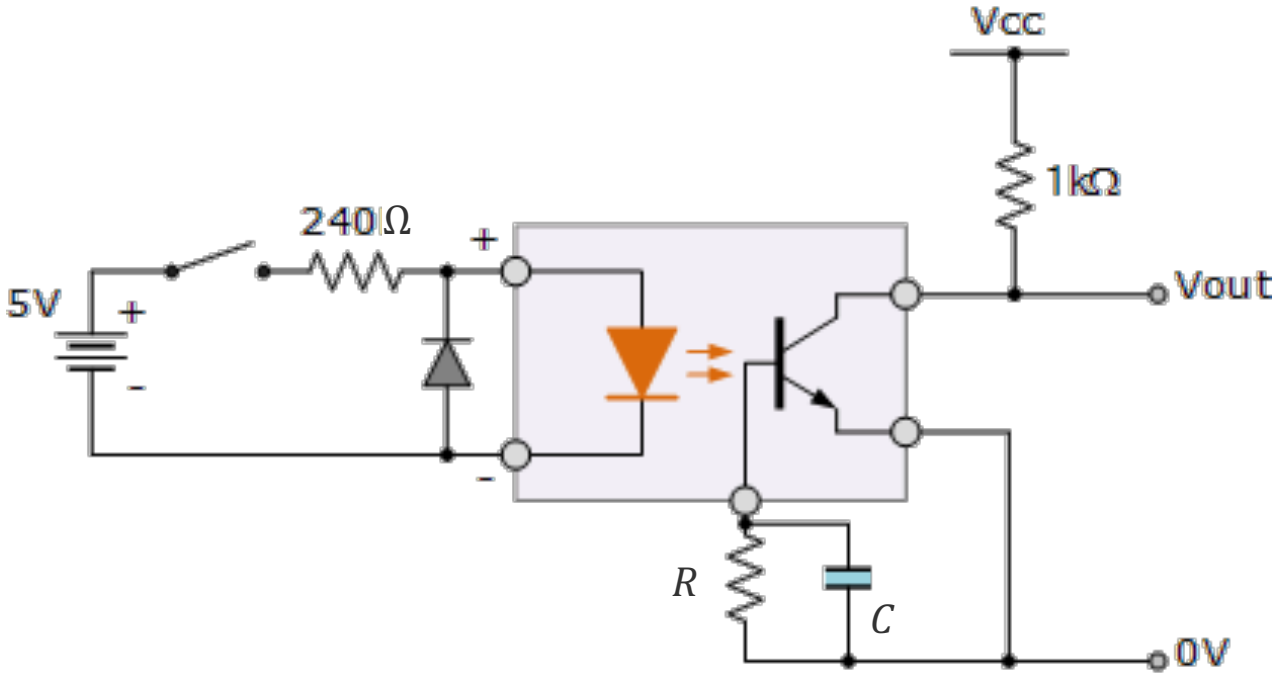
Zero-crossing Detector

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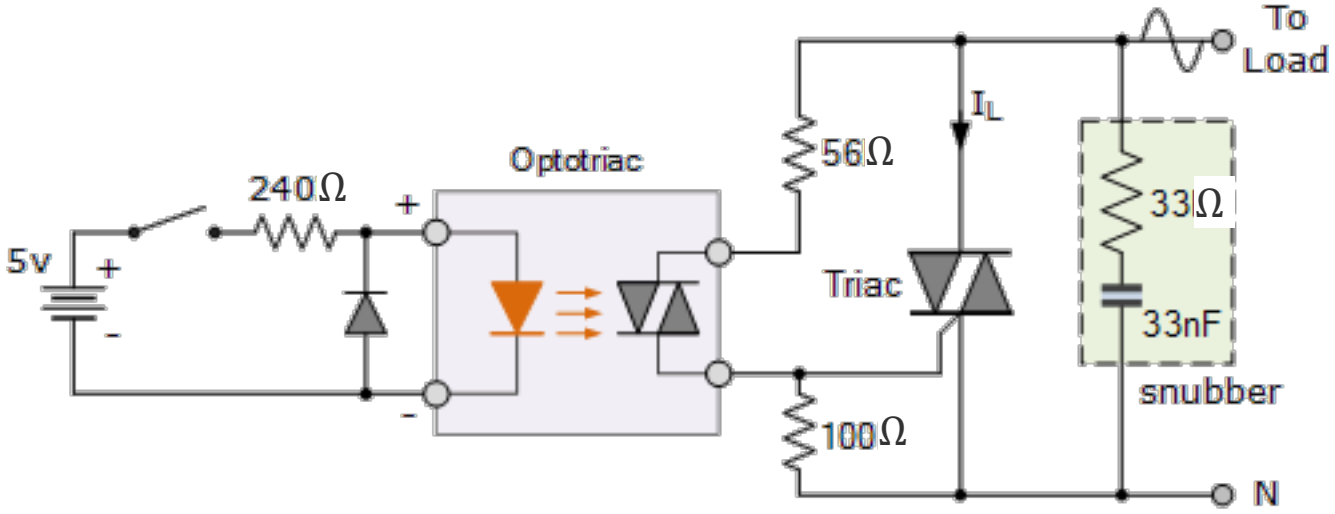
A DC Switch

The value of the resistor *R* can be chosen to suit the selected photo-coupler device and the amount of switching sensitivity required. The capacitor stops any unwanted spikes or transients from false triggering the opto-transistors base.



Simple Solid-state Relay

Opto-coupled triacs such as the MOC 3020, have voltage ratings of about 400 volts making them ideal for direct mains connection and a maximum current of about 100 mA. For higher powered loads, the opto-triac may be used to provide the gate pulse to another larger triac via a current limiting resistor.



Q & A

Send me an email if you have any questions.